

A gap in the proof of Lemma 3.12 of  
Cavenagh–Greenhill–Wanless,  
*The cycle structure of two rows in a random Latin square*

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**Abstract**

Case 3 of the splitting procedure used to prove Lemma 3.12 of [1] (“cross-switch with respect to  $\{\omega(j), \omega(j')\}$ , then backflip with respect to  $\{j, j'\}$ ”) does not produce a Latin square of the intended type when  $\min(\alpha, \beta) \geq 3$ : the cross-switch reverses an  $\omega$ -arc containing  $j'$ , after which  $\{j, j'\}$  is at  $\omega$ -distance 2. The lower bound of Lemma 3.12 survives (and is in fact elementary); the upper bound  $3/2$ , hence the upper bound of Theorem 3.13 and everything downstream of it, is not established by the published argument for those  $(\alpha, \beta)$ . A canonical repair exists when  $\alpha$  is even and  $\alpha < 2\beta$  (or symmetrically); we do not see one when  $\alpha, \beta$  are both odd. Exact data for  $n \leq 11$  and sampled squares up to  $n = 100$  indicate that the statement is true with much room to spare.

## 1 Setting and the procedure

We use the notation of [1], Section 3.  $L$  is a Latin square of order  $n$  containing a suitable partial square  $F$ ;  $\omega$  is the permutation of the first  $m$  columns with  $1 \circ \omega(j) = 2 \circ j$ , whose cycle type  $\lambda \in P(m)$  defines  $S(\lambda, F)$ . For columns  $j, j'$  let  $\delta_{j,j'}$  be the row permutation with  $\delta(i) \circ j = i \circ j'$ ;  $\{j, j'\}$  is an *A-pair* if rows 1, 2 lie in different cycles of  $\delta_{j,j'}$  and a *B-pair* otherwise. The *backflip* at an *A-pair*  $\{j, j'\}$  lying on one  $\omega$ -cycle of length  $\alpha + \beta$  with  $\omega^\alpha(j) = j'$  trades the  $\delta_{j,j'}$ -cycle through row 2; it splits that  $\omega$ -cycle into the two arcs between  $j$  and  $j'$ , of lengths  $\alpha$  and  $\beta$  (Lemma 3.5). The *cross-switch* at a *B-pair*  $\{j, j'\}$  with  $\omega^\alpha(j) = j'$  (Definition 3.9) is a Latin bitrade  $(Q, Q')$  that preserves  $S(\lambda, F)$  (Lemma 3.10).

Let  $\mu$  be  $\lambda$  with one part  $\alpha + \beta$  split into  $\alpha, \beta \geq 2$ . The splitting procedure (p. 294) considers, for  $L \in S(\lambda, F)$ , every pair  $\{j, j'\}$  with  $\omega^\alpha(j) = j'$  on an  $(\alpha + \beta)$ -cycle and produces  $L' \in S(\mu, F)$  as follows:

1. if  $\{j, j'\}$  is an *A-pair*, backflip at  $\{j, j'\}$ ;
2. else if  $\{\omega j, \omega j'\}$  is an *A-pair*, backflip at  $\{\omega j, \omega j'\}$ ;
3. otherwise cross-switch with respect to  $\{\omega j, \omega j'\}$ ; “now  $\{j, j'\}$  has become an *A-pair*, and we backflip with respect to  $\{j, j'\}$ ”.

Item 4 asserts that in each case  $L' \in S(\mu, F)$ . Together with a fifth edge (a switch) this gives every  $L \in S(\lambda, F)$  degree exactly  $2(\mu_{\alpha+\beta} + 1)(\alpha + \beta)$  in a bipartite multigraph between  $S(\lambda, F)$  and  $S(\mu, F)$ , while the joining procedure gives every  $L \in S(\mu, F)$  degree between  $\mu_\alpha \mu_\beta \alpha \beta$

and three times that; the two multigraphs are asserted to coincide, and the double count is Lemma 3.12:

$$\frac{1}{2} \leq \frac{|S(\lambda, F)| (\mu_{\alpha+\beta} + 1)(\alpha + \beta)}{|S(\mu, F)| \mu_{\alpha} \mu_{\beta} \alpha \beta} \leq \frac{3}{2} \quad (\alpha \neq \beta). \quad (1)$$

## 2 The cross-switch reverses an arc

Write  $e_1 = 1 \circ j$ ,  $e_2 = 1 \circ j'$ ,  $e_3 = 2 \circ j$ ,  $e_4 = 2 \circ j'$  and consider the first form of Definition 3.9 ( $\min(e_1, e_2, e_3, e_4) \in \{e_1, e_4\}$ ); the second form is the mirror image with the roles of  $(j, \text{row } 1)$  and  $(j', \text{row } 2)$  exchanged. On rows 1, 2 the bitrade is

$$Q' \supseteq \{(2, j, e_2), (1, j', e_3)\} \cup \{(1, \omega^k j, 2 \circ \omega^k j), (2, \omega^k j, 1 \circ \omega^k j) : 1 \leq k < \alpha\},$$

i.e. in every interior column of the  $\omega$ -arc from  $j$  to  $j'$  the entries of rows 1, 2 are swapped, row 2 receives  $e_2 = 1 \circ j'$  at column  $j$ , and row 1 receives  $e_3 = 2 \circ j$  at column  $j'$ . (This is forced by the proof of Lemma 3.10: row 1 must keep the symbol set  $\{1 \circ \omega^k j : 1 \leq k \leq \alpha\}$  on the cells  $\{\omega^k j : 1 \leq k < \alpha\} \cup \{j'\}$ , and “precisely one column from the pair  $\{\omega^k j, \omega^{k'} j'\}$  has symbols swapped in rows 1 and 2”.) The remaining cells of  $Q$  lie in columns  $j, j'$  and in rows other than 1, 2; they do not affect  $\omega$ .

**Lemma 1.** *Let  $L'$  be the cross-switch of  $L$  at the  $B$ -pair  $\{j, j'\}$ ,  $\omega^\alpha(j) = j'$ , and let  $\omega' = \omega_{L'}$ . Then  $\omega'(j) = \omega^{\alpha-1}(j)$ ,  $\omega'(\omega^k j) = \omega^{k-1}(j)$  for  $2 \leq k < \alpha$ ,  $\omega'(\omega j) = j'$ , and  $\omega' = \omega$  on every other column. In words: the interior of the  $\omega$ -arc from  $j$  to  $j'$  is traversed in reverse, and the cycle structure of  $\omega$  is otherwise unchanged.*

*Proof.* Write  $1', 2'$  for the new rows. Then  $1' \circ j = e_1$ ,  $1' \circ \omega^k j = 1 \circ \omega^{k+1} j$  for  $1 \leq k < \alpha$  (since  $2 \circ \omega^k j = 1 \circ \omega^{k+1} j$ ),  $1' \circ j' = 1 \circ \omega j$ ;  $2' \circ j = 1 \circ j' = 1 \circ \omega^\alpha j$ ,  $2' \circ \omega^k j = 1 \circ \omega^k j$  for  $1 \leq k < \alpha$ ,  $2' \circ j' = e_4$ . Now  $\omega'(c)$  is the column whose row-1' entry equals  $2' \circ c$ : for  $c = j$  this is  $\omega^{\alpha-1} j$ ; for  $c = \omega^k j$  with  $2 \leq k < \alpha$  it is  $\omega^{k-1} j$ ; for  $c = \omega j$  it is  $j'$  (the symbol  $1 \circ \omega j$  now sits at  $(1', j')$ ); for  $c = j'$  it is  $\omega j'$ , unchanged; for  $c = \omega^{-1} j$ ,  $2' \circ c = 2 \circ c = e_1 = 1' \circ j$ , so  $\omega'(c) = j$ ; all other columns are untouched in both rows.  $\square$

The example on p. 293 of [1] shows exactly this: there  $\omega = (1234567)$ ,  $j = 2$ ,  $j' = 5$ ,  $\alpha = 3$ , and the displayed cross-switch has rows 1245367 and 2534671, i.e.  $\omega' = (1243567)$ .

## 3 Case 3 lands in the wrong type

**Proposition 2.** *Let  $\{j, j'\}$  and  $\{J, J'\} = \{\omega j, \omega j'\}$  both be  $B$ -pairs with  $\omega^\alpha j = j'$  on an  $(\alpha + \beta)$ -cycle, and let  $L'$  be the cross-switch of  $L$  at  $\{J, J'\}$  (either form). Then  $\{j, j'\}$  is an  $A$ -pair of  $L'$ , but in the first form it is at  $\omega'$ -distance 2 (from  $j$  to  $j'$ ), and in the second form at  $\omega'$ -distance 2 from  $j'$  to  $j$ . Consequently the backflip at  $\{j, j'\}$  splits the  $(\alpha + \beta)$ -cycle into parts 2 and  $\alpha + \beta - 2$ , which is the type  $\mu$  only if  $\min(\alpha, \beta) = 2$ .*

*Proof.* First form. The arc from  $J = \omega j$  to  $J' = \omega^\alpha J$  has interior  $\omega J, \dots, \omega^{\alpha-1} J$ , and  $\omega^{\alpha-1} J = \omega^\alpha j = j'$ . By Lemma 1,  $\omega'(j) = \omega'(\omega^{-1} J) = J$  and  $\omega'(J) = \omega^{\alpha-1} J = j'$ . The status of  $\{j, j'\}$ : column  $j'$  is an interior column of the arc, so rows 1, 2 are swapped there, while column  $j$  is untouched ( $j \notin \{J, J'\}$  and  $j$  is not in the arc); hence  $\delta_{j, j'}$  changes by a transposition of rows 1, 2 and  $\{j, j'\}$  toggles from  $B$  to  $A$ , as claimed in [1]. The second form reverses the arc from  $J'$  to  $J$ , whose interior contains  $j = \omega^{-1} J$  as its last column, and the same computation gives  $\omega'(J') = j$ , i.e.  $\omega'^2(j') = j$ .  $\square$

So the asserted membership  $L' \in S(\mu, F)$  in item 4 fails in case 3 whenever  $\min(\alpha, \beta) \geq 3$ , and the joining rule 4 (cross-switch at  $\{\omega j, \omega j'\}$  after a flip) fails symmetrically, so the two multigraphs do not coincide. For  $\min(\alpha, \beta) = 2$  the arc has a single interior column, nothing is reversed, and the proof is complete; this is the case of the sharp example  $n = 5$ .

**Numerical confirmation.** We implemented Definition 3.9 literally (rows 1, 2 as above; in columns  $j, j'$  the rows  $\delta^k(1)$ ,  $1 \leq k < a$ , with  $\delta^a(1) = 2$ , swap their two entries, exactly as in the definition — this is what balances the corner cells) and checked on uniformly sampled Latin squares (Jacobson–Matthews) that the cross-switch is a Latin square of the same type in every instance. Applying case 3 as written: at  $n = 9$  and  $n = 12$ , in 1006 and 1670 instances with  $\min(\alpha, \beta) \geq 3$  respectively, the result had the wrong type in every single instance (e.g.  $(2, 7)$  with  $(\alpha, \beta) = (3, 4)$  gives  $(2, 2, 5)$  instead of  $(2, 3, 4)$ ;  $(9)$  with  $(3, 6)$  gives  $(2, 7)$ ).

## 4 What survives, and what depends on the gap

Only item 4's claim in case 3 is affected. Cases 1 and 2 and the switch edge are valid, so each  $L \in S(\lambda, F)$  has degree at most  $2(\mu_{\alpha+\beta} + 1)(\alpha + \beta)$  in the multigraph of valid edges, and each  $L \in S(\mu, F)$  has degree at least  $\mu_\alpha \mu_\beta \alpha \beta$  (joining rules 1–2 are valid). Hence the *lower* bound  $\frac{1}{2}$  in (1) stands. (In fact it is elementary: with  $X$  the set of pairs  $(L, \{j, j'\})$ ,  $L \in S(\lambda, F)$ ,  $\omega^\alpha j = j'$ , and  $J$  the analogous set of joining pairs, the flip and the switch give  $|J| = 2|X_A|$  exactly, so (1) reads  $\frac{1}{3} \leq |X_A|/|X| \leq 1$ , and the lower bound  $\frac{1}{2}$  is  $|X| \geq |X_A|$ .) The *upper* bound  $\frac{3}{2}$ , equivalently

$$\Pr[\{j, j'\} \text{ is an } A\text{-pair}] \geq \frac{1}{3} \quad \text{for } (L, \{j, j'\}) \text{ uniform on } X,$$

is not established for  $\min(\alpha, \beta) \geq 3$ .

Within [1] the upper bound of Lemma 3.12 is used for: the upper bound of Theorem 3.13; the lower bound  $(2/3)^{\kappa-1}$  of Lemma 4.1 and the corresponding half of Lemma 4.2; the parity bounds of Theorem 4.3 (whose proof invokes the equality of the two multigraphs “as in the proof of Lemma 3.12”); the upper bound  $\mathbb{P}_F((m)) \leq 2m^{-2/3}$  of Lemma 4.4 and hence Corollary 4.5 (the factor  $n^{1/3}$ ); and those results of Section 4.2 that rest on these. The upper bound  $2^{\kappa-1}$  of Lemma 4.1, the lower bound  $\mathbb{P}_F((m)) \geq 2m^{-2}$ , and the upper bounds on short-cycle counts derived from the joining direction are unaffected. For instance, from the joining direction alone one still gets  $\mathbb{E}[\ell N_\ell] \leq 2m/(m - \ell)$  for the number  $N_\ell$  of  $\ell$ -cycles (sum the inequality  $\mathbb{E}[\ell N_\ell \beta N_\beta] \leq 2\mathbb{E}[(\ell + \beta)N_{\ell+\beta}]$  over  $\beta$ ), and by joining  $k$  cycles successively one obtains  $\mathbb{E}[(\kappa)_k] \leq 2^{k-1}(4 + k^2/\log m)(\log m)^k$  for the number of cycles  $\kappa$ , hence  $\Pr[\kappa \geq 16 \log m] = O(m^{-2} \log m)$ ; so the tail of the number of cycles survives, while the lower bound on the number of cycles in Section 4.2 does not.

## 5 A partial repair

A cross-switch at a  $B$ -pair  $\{J, J'\}$  at any  $\omega$ -distance  $a \geq 2$  is a type-preserving trade (the proof of Lemma 3.10 does not use  $a = \alpha$ ), and by Lemma 1 it reverses the arc from  $J$  to  $J'$ . It toggles the status of  $\{j, j'\}$  iff exactly one of  $j, j'$  is an interior column of that arc, say at position  $i$ , and then the  $\omega$ -distance from  $j$  to  $j'$  changes by  $a - 2i$ . Thus the status-toggling moves that preserve the distance are precisely the cross-switches at  $B$ -pairs *symmetric about  $j$  or about  $j'$*  ( $a = 2i$ ).

Suppose  $\alpha$  is even and  $\alpha < 2\beta$ , and let  $R = \{\omega^{\alpha/2}j, \omega^{3\alpha/2}j\}$ : the pair at distance  $\alpha$  symmetric about  $j'$ , whose arc avoids  $j$ . If  $R$  is a  $B$ -pair, cross-switch at  $R$ :  $\{j, j'\}$  becomes an  $A$ -pair at distance  $\alpha$ , and the backflip at  $\{j, j'\}$  lies in  $S(\mu, F)$ . If  $R$  is an  $A$ -pair, backflip at  $R$  itself, which is a pair at distance  $\alpha$ . Either way case 3 acquires a valid edge (verified on 3928 sampled instances at  $n = 10, 13$ : type  $\mu$  in all of them). The same works with the roles of  $\alpha, \beta$  exchanged when  $\beta$  is even and  $\beta < 2\alpha$ . The multiplicities on the  $S(\mu, F)$  side change (an  $A$ -pair's backflip may now serve two splitting pairs), so the constant  $3/2$  would have to be recomputed.

When  $\alpha$  and  $\beta$  are both odd, every distance-preserving toggle has even arc length, so no symmetric pair is at distance  $\alpha$  or  $\beta$ ; if all symmetric pairs about  $j$  and  $j'$  happen to be  $A$ -pairs, no single cross-switch repairs  $\{j, j'\}$ , and two-step repairs pass through a third type, which breaks the two-sided count. We do not see a canonical repair in this case (nor when  $\alpha \geq 2\beta$  with  $\beta$  odd). Repairs that use other rows do exist — for a  $B$ -pair  $\{j, j'\}$  with column cycle  $Z \ni 1, 2$ , trade rows  $z, w$  lying on different arcs of  $Z$  between 1 and 2 along the row cycle of  $\{z, w\}$  through  $j$  (if  $j, j'$  lie in different row cycles of  $\{z, w\}$ ; this separates 1 from 2 and keeps the type), or cross-switch rows  $z, w$  at  $\{j, j'\}$  (if they lie in one; this changes the type to  $\mu$ ) — but they are indexed by a free pair of rows and so have unbounded multiplicity on the  $S(\mu, F)$  side, which defeats the double count.

## 6 What the data say

The statement of Lemma 3.12 is almost certainly true, and far from sharp for  $n \geq 6$ . By the identity  $|J| = 2|X_A|$  the quantity in (1) is  $\mathcal{C}_n(\lambda)/\mathcal{C}_n(\mu)$ , where  $\mathcal{C}_n(\lambda)$  is the number of completions of two rows of type  $\lambda$ , and  $\Pr_X[A] = \mathcal{C}_n(\mu)/(2\mathcal{C}_n(\lambda))$ . From the exact values of  $\mathcal{C}_n(\lambda)$  tabulated in [1] (Tables 1 and 2,  $n \leq 11$ ):

| $n$ | #splits $(\lambda \rightarrow \mu)$ | $\max \mathcal{C}_n(\lambda)/\mathcal{C}_n(\mu)$ | $\min \Pr_X[A]$ |
|-----|-------------------------------------|--------------------------------------------------|-----------------|
| 5   | 1                                   | 1.5000 (sharp)                                   | 0.3333          |
| 6   | 3                                   | 0.9545                                           | 0.5238          |
| 7   | 4                                   | 1.0035                                           | 0.4982          |
| 8   | 8                                   | 1.0017                                           | 0.4991          |
| 9   | 11                                  | 1.0002                                           | 0.4999          |
| 10  | 19                                  | 1.0000                                           | 0.5000          |
| 11  | 26                                  | 1.0000                                           | 0.5000          |

(all splits with  $\alpha, \beta \geq 2$ ; for  $n \geq 7$  every ratio is within 0.4% of 1, in the covered and the uncovered classes alike — e.g.  $(2, 6) \rightarrow (2, 3, 3)$  at  $n = 8$ , in the  $(3, 3)$  class, has ratio 1.0017). On uniformly sampled squares with a uniformly random marked pair at distance  $\alpha \geq 2$  on an  $m$ -cycle,  $m - \alpha \geq 2$ , we find  $\Pr_X[B] - \Pr_X[A] = -0.022 \pm 0.011, +0.008 \pm 0.012, -0.030 \pm 0.018$  at  $n = 30, 50, 100$  (9000, 7200, 3600 instances), i.e.  $\Pr_X[A] = 0.50 \pm 0.01$  throughout. So the true constant in the upper bound of (1) appears to be  $1 + O(n^{-2})$  rather than  $3/2$ ; the question is only what has been proved.

## References

- [1] N. J. Cavenagh, C. Greenhill and I. M. Wanless, The cycle structure of two rows in a random Latin square, *Random Structures & Algorithms* 33 (2008), 286–309.